VLSM Subnetting (Address Block method)

Rules: Consider subnet mask and deternine the number of host bits. If the number of host bits is *n*, the host address space,

- ***(i). is a block of fixed size 2^n***
- ***(ii). m start at fixed boundaries i.e., $k2^n$ where** $k = 0, 1, 2, ...$*
- ***(iii). must not overlap with other used spaces.***

Notes:

1. The number of host bits determine the host address space, e.g. 8 bits gives 2^8=256 possible addresses, 7 bits gives 2^7=128 possible addresses, etc.
2. The first address in the block is the network address. The host bits are all zeros.
3. The last address in the block is the (directed) broadcast address. The host bits are all ones.
4. The host bit can occur in any octet. If the subnet masks are 8, 16, or 24, the host bits are 24, 16, and 8 respectively.
5. If first host bit occurs inside an octet, (called the boundary octet), e.g. if the mask is 20, the boundary octet is the 3rd octet. Consider only the host bits in this octet.

I. Determining network address

Consider the "boundary octet" where the host part of the IP address starts. Determine the number of host bits *n* in this octet. The address block size is 2^n. The network addresses are $k2^n$ where $k = 0,1,2$ ..., such that $k2^n < 256$.

Example: What is the network address of this host 182.24.57.75/20?
Solution:
The host part occurs in the 3nd octet, i.e. the boundary octect. There are host 4 bits in this octet. The host addresses in this octet are multiples of 2^4, i.e. 0, 16, 32, 48, 64, ..., 240. Hence the above address belongs to the 182.24.48.0/20 subnet.

Example 1: Given address `202.56.64.134/27,` what is the network and broadcast address?

Solution

The subnet mask boundary occurs in the 4th octet. Consider only this octet.
Step 1. How many host bits are there? Number of host bits = 5
hence, address block size: $2^5 = 32$ addresses
Step 2. What is the starting addresses for the address blocks in this boundary octet?
Possible starting addresses in the 4th octet are:
0, 32, 64, 96, 128, 160, 192, & 224.
Step 3. Locate the address block where the given address is located, i.e. 128 - 159

Ans: Network address: 202.56.64.128 (first address)
Broadcast address: 202.56.64.159 (last address)

Example 2: Given address `203.45.156.148/18`, what is the network and broadcast address?
<u>Solution</u>
Consider only the 3^{rd} octet, where the subnet mask (18 bits) boundary occurs.
Step 1: How many host bits in this octet? number = 24 - 18 = 6 bits
hence, address block size in this octet: 2^6 = 64 addresses.
Step 2: What are the starting addresses for the address blocks?
Possible address blocks in the 3^{rd} octet start at
0, 64, 128, & 192

Ans: For the give address 203.45.156.148/18
Network address: 203.45.128.0 (first address)
Broadcast address: 203.45.191.0 (last address)

II. Designing Subnets

Example1 : Given a network `192.168.2.0/24`, create **Subnet A** with 60 hosts, **Subnet B** has 20 hosts and **Subnet C** with 12 hosts.

<u>Solution</u>
Note: It is easiest to start with the biggest size subnet, working down to the smallest.
<u>*Subnet A (60 hosts):*</u>
Step 1: Determine address block size:
how many host bits required: 6 bits, giving 2^6 = 64 > 60, prefix /26
address block size available: 2^6 = 64 addresses

Step 2: Determine possible starting addresses: 0, 64, 128, 192
choose 0 as the starting address,
hence network address: 192.168.2.0/26
1^{st} useable address 192.168.2.1
last useable address 192.168.2.62
broadcast address: 192.168.2.63
<u>*Subnet B (20 hosts):*</u>
Step 1: Determine address block size:
how many host bits required: 5 bits, gives 2^5=32 >20, prefix /27
address block size available: 2^5 = 32 addresses

Step 2: Determine possible starting addresses: 0, 32, 64, 96, 128, 160, 192,224
Choose starting address, Since 0 to 63 were taken by A, choose 64
hence, network address: 192.168.2.64/27
1^{st} useable address 192.168.2.65
last useable addres 192.168.2.94
broadcast address: 192.168.2.95

Subnet C (12 hosts):
Step 1: Determine address block size:
how many host bits required: 4 bits, gives 2^4=16>12, hence prefix is /28
address block size available: 2^4 = 16 addresses

Step 2: Determine possible starting addresses: 0, 16, 32, 48, 64, 80, 96, 112,128, ...

Choose starting address, Since 64-95 were taken by B, choose 96

hence, network address: 192.168.2.96/28

1st useable address 192.168.2.97

last useable addres 192.168.2.110

broadcast address: 192.168.2.111

Example 2: Given subnet `203.45.128.0/18`, create **Subnet A** having subnet prefix /22 and **Subnet B** having subnet prefix /25.

<u>*Subnet A:*</u>

Step 1: Prefix /22 occurs in the 3rd octet,

how many hosts bits available there: 24 - 22 = 2 bits,

address block size: 2^2 *= 4 addresses*

Step 2: Determine starting address in 3rd octet: 0,4,8,12,16, ...,128, 132, ...

Possible starting address are 128, 132 since we are given `203.45.128.0/18` *to use*

Choose the 1st address block

hence, network address: 203.45.128.0/22

1st useable address: 203.45.128.1

last useable address: 203.45.131.254

broadcast address: 203.45.131.255

<u>*Subnet B:*</u>

Step 1: Prefix /25 occurs in 4th octet,

how many hosts bits available there? 32 - 25 = 7 bits,

address block size: 2^7 *= 128*

Step 2: Determine starting address in 4th octet: 0, 128

Possible starting addresses are: 0, 128, let's choose 0, to be contiguous with Subnet A.

hence, network address: 203.45.132.0

1st useable address: 203.45.132.1

last useable address: 203.45.132.126

broadcast address: 203.45.132.127